

4COSC002W Mathematics for Computing

Lecture 2

Set Theory Basics. Operations with Sets. Intervals

UNIVERSITY OF
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Introduction to Set Theory

Set Theory Basics

What is a Set?

Sets are fundamental objects in mathematics, defined as collections of distinct *elements*.

The elements that make up a set can be anything: *numbers, letters, other sets, and so on*.

Sets are typically denoted using *curly braces*, with elements separated by commas.

Examples: $\{a,b,c\}$ $\{\odot, 5, \clubsuit\}$ $\{1, 2, 3, 5, 7\}$

Sequences vs Sets

Property	Sets	Sequences
Definition	A collection of distinct objects.	An ordered list of elements, possibly with repetition.
Ordering	Order of elements is not significant.	Order of elements is significant.
Repetition	No repetition of elements is allowed.	Repetition of elements is allowed.
Notation	Typically denoted using curly braces {}.	Represented using parentheses (), angle brackets <>, or listed.
Membership	An element either belongs or does not belong.	An element can occur multiple times at different positions.
Indexing	Elements are not indexed.	Elements are indexed; there is a first element, a second element, etc.
Cardinality/Length	Cardinality is the number of distinct elements.	Length is the total number of elements, including repetitions.

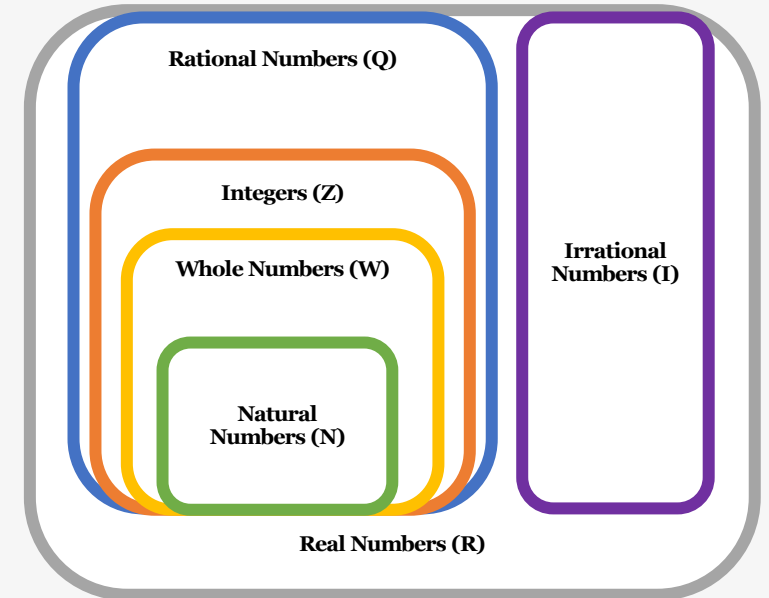
Some well-known and named sets

“N” is the natural numbers $\{0, 1, 2, 3, 4, 5, \dots\}$

“Z” is the set of integers $\{\dots-2, -1, 0, 1, 2, \dots\}$

“Q” is the set of rational numbers

“R” is the set of real numbers



Empty Set. Cardinality

There are some special sets, such as the **empty set** $\emptyset$ or $\{\}$, which has no elements, and **universal sets** (U), which contain all elements under consideration.

The **cardinality of a set** is the number of elements in the set. It is denoted as $|A|$ for a set A .

NB: If the *empty sets* are included on purpose as the elements of the sets, they are counted as separate elements and are included in set cardinality.

ACTIVITY 1: Solve the following problems

$$A = \{\textit{alpha}, \textit{beta}, \textit{gamma}\}$$

$$B = \{-5, 0, 5, 10\}$$

$$C = \{\{a, \{b\}\}, \{c, d\}\}$$

$$D = \{\{\}, \{\}, 10, 11\}$$

$$E = \{\}$$

1. What is $|A|$, $|D|$?

$$|A|=3, |D|=3$$

2. What is $|B|+|C|$

$$6$$

3. What is $|D|+|E|-|A|$?

$$0$$

ACTIVITY 1 DETAILED ANSWER

- **What is $|A|$, $|D|$?**

1. Set $A = \{\alpha, \beta, \gamma\}$. There are 3 elements, so $|A| = 3$.
2. Set $D = \{\{\}, \{\}, 10, 11\}$. The empty set $\{\}$ is counted as an element itself, so there are 3 unique elements: $\{\}$, 10, and 11. Thus, $|D| = 3$.

The answer is: $|A| = 3$, $|D| = 3$.

- **What is $|B| + |C|$?**

1. Set $B = \{-5, 0, 5, 10\}$. There are 4 elements, so $|B| = 4$.
2. Set $C = \{\{a, \{b\}\}, \{c, d\}\}$. There are 2 elements, which are $\{a, \{b\}\}$ and $\{c, d\}$, so $|C| = 2$.

Therefore, $|B| + |C| = 4 + 2 = 6$.

- **What is $|D| + |E| - |A|$?**

1. $|D| = 3$, as determined earlier.
2. Set $E = \{\}$ is the empty set, so $|E| = 0$.
3. $|A| = 3$, as determined earlier.

Therefore, $|D| + |E| - |A| = 3 + 0 - 3 = 0$.

Equal sets

A set A is **equivalent** to a set B (abbreviated as $A \equiv B$) if the following conditions hold

- 1) Every element of A is an element of B
- 2) Every element of B is an element of A

Examples:

$\{1,2,3\} \equiv \{1,2,3\}$ as both conditions above hold

$\{\clubsuit, \odot, 5\} \equiv \{\odot, 5, \clubsuit\}$

Subsets

A **set** A is considered a **(weak) subset of set B** if every element of set A is also an element of set B .

The notation used to represent this relationship is $A \subseteq B$, which can be read as "A is a subset of B".

This is analogous to the relation "less or equal."

The empty set $\emptyset$ is a subset of every set.

Examples:

- If $A = \{1, 2\}$ and $B = \{1, 2, 3, 4\}$, then $A \subseteq B$, because every element of set A (which are 1 and 2) is also an element of set B .
- If $C = \{1, 2, 3, 4\}$, then $C \subseteq B$, also $C \equiv B$.

Proper Subset

If A is a subset of B , and A is not equal to B (meaning some element in B is not in A), then A is called a **proper subset of B** .

The notation used for a proper subset is $A \subset B$.

This is analogous to the relation “less”.

Examples:

- If $A = \{a, b\}$ and $B = \{a, b, c, d, e\}$, then $A \subset B$ because A is a subset of B , and there exist the elements in B (which are c, d, e) that are not in A . If $C = \{a, b, c, d, e\}$, it is NOT a proper subset of B .
- Remember, by this definition, every set is considered a subset of itself ($A \subseteq A$) since all elements of A are indeed in A . However, a set is not a proper subset of itself.

Set Builder Notation

Set builder notation is a shorthand way to describe a set by indicating the properties that its members must satisfy.

The general format for set builder notation:

$\{\text{variable} \mid \text{property or condition}; \text{property or condition 2}, \dots\}$

The **variable** is a placeholder that represents the elements of the set.

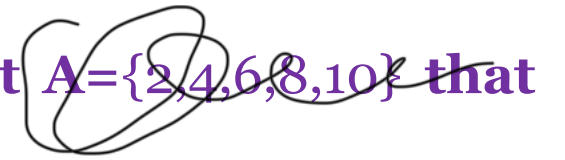
The **property or condition** specifies the criteria elements must satisfy to be included in the set.

The vertical bar (or colon) can be read as "such that".

Set Builder Notation Examples

Example 1: A subset should contain numbers from the set $A = \{2, 4, 6, 8, 10\}$ that are divisible by 4

Set Builder Notation: $\{x \mid x \in A; x \bmod 4 = 0\}$; Result: $\{4, 8\}$



$x \in N$

Example 2: A subset should contain objects from the set $\{a, b, c, d, e\}$ that come before 'd' alphabetically:

Description: The set of alphabets from the given subset that come before the letter 'd' alphabetically.

Set Builder Notation: $\{x \mid x \in \{a, b, c, d, e\} \text{ and } x < d\}$; Result: $\{a, b, c\}$

$x < d$

Example 3. A subset should contain numbers from the set $\{1, 2, 3, 4, 5\}$ that are greater than 3:

Description: The set of numbers greater than 3 from the given set.

Set Builder Notation: $\{x \mid x \in \{1, 2, 3, 4, 5\}; x > 3\}$; Result: $\{4, 5\}$



ACTIVITY 2:

Set Builder Notation with Tuples

Consider two sets: $A=\{1,2,3\}$ and $B=\{a,b\}$

1. Define the set S using set builder notation that represents all possible ordered pairs, *called tuples, (x,y)* where x belongs to set A and y belongs to set B .
2. Define a set T using set builder notation, which consists of ordered pairs from S such that the first element of the tuple is greater than 2.

ACTIVITY 2 Answer

Question 1.

$$S = \{(x, y) \mid x \in A, y \in B\}$$

Explicitly, $S = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$

Question 2.

$$T = \{(x, y) \mid x \in A, y \in B, x > 2\}$$

Explicitly, $T = \{(3, a), (3, b)\}$

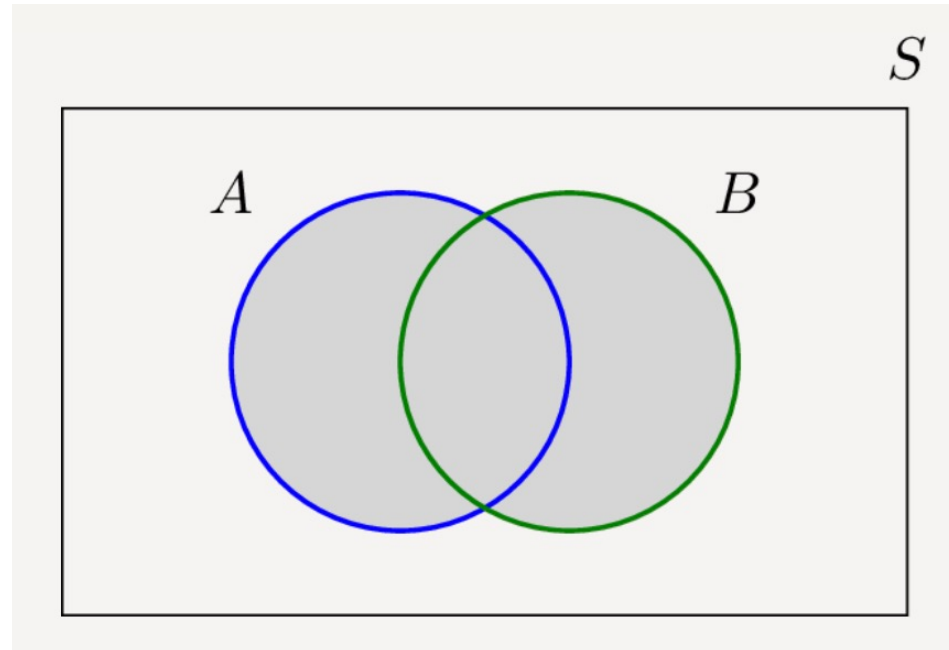
Operations with Sets

Set Theory Basics

Set Operations: Union

$$A \cup B$$

Create a new set by combining *all of the elements* of two sets
All elements that are in A or in B.



Venn Diagram

Union Examples

*Given**

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

$$C = \{6, 7, 8\}$$

$$D = \{\} \text{ (Empty set)}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

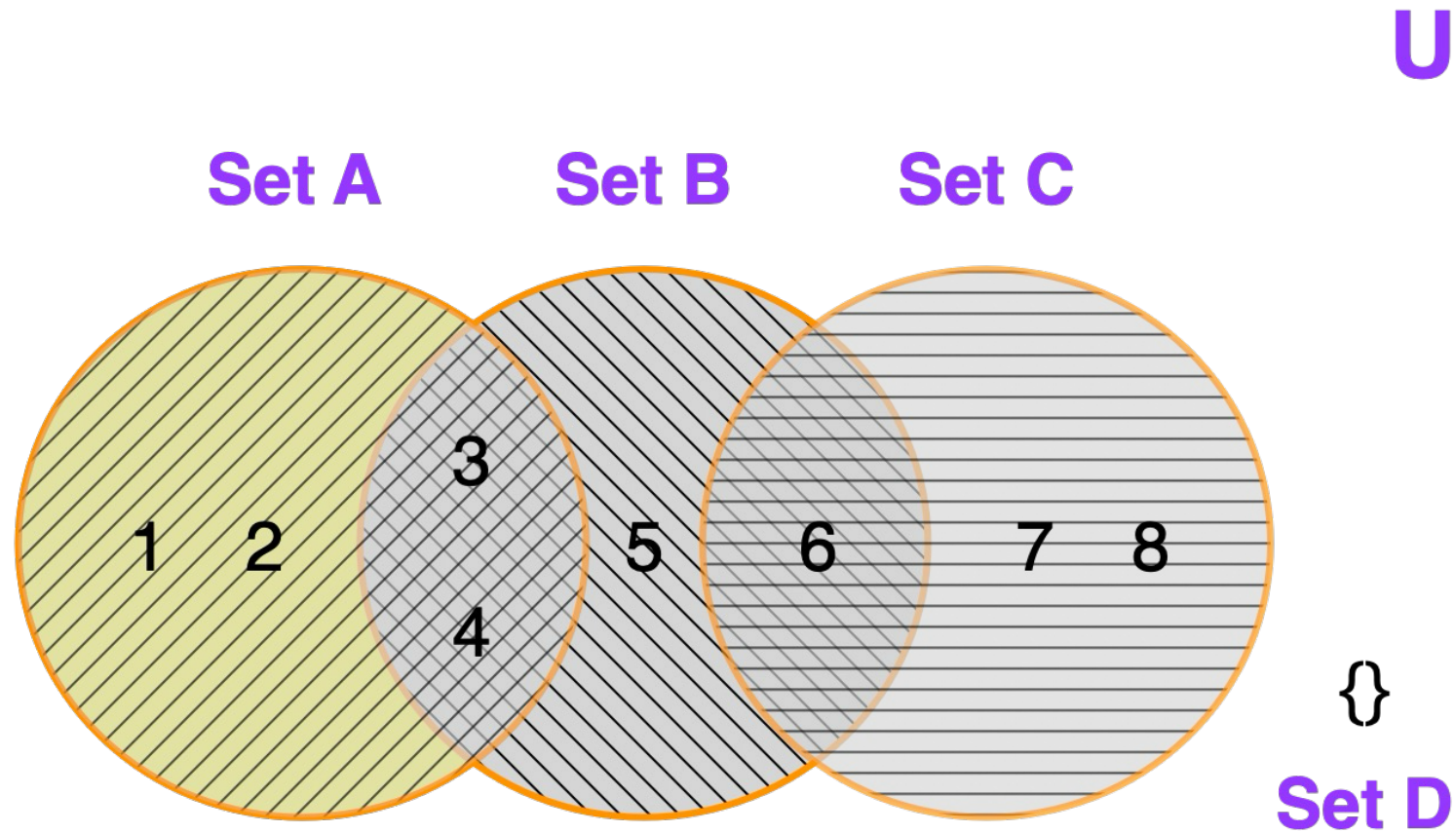
$$A \cup D = \{1, 2, 3, 4\}$$

(union with an empty set
results in the non-empty set)

$$B \cup C = \{3, 4, 5, 6, 7, 8\}$$

*See the diagram on the next slide

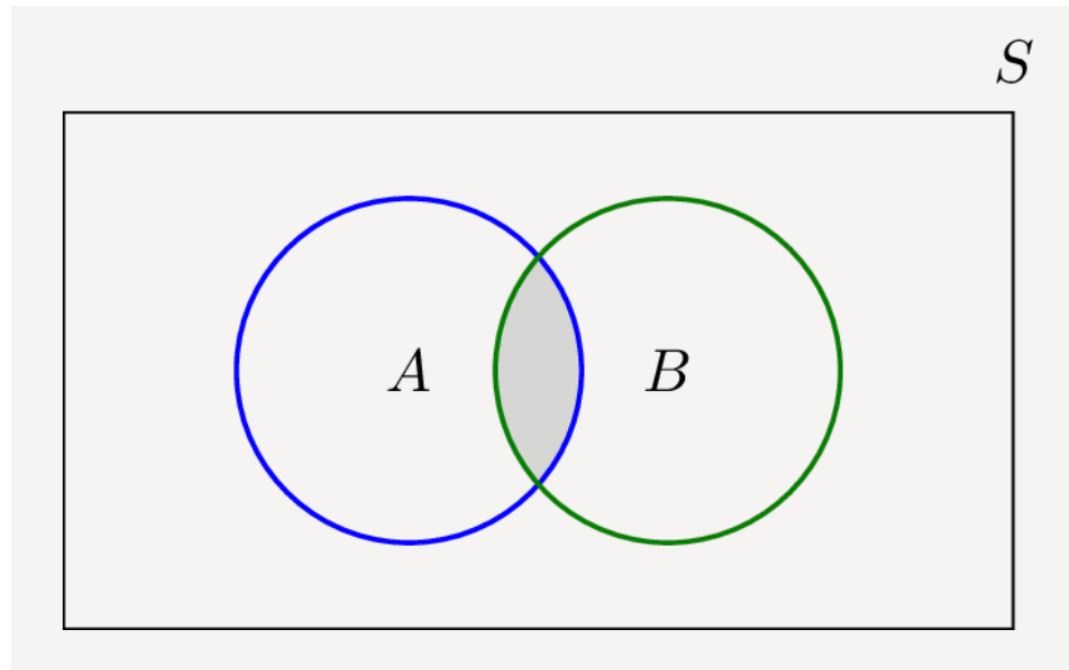
Sets Operations Example Venn Diagram



Set Operations: Intersection

$$A \cap B$$

Create a new set (C), using the elements the two sets have *in common*.
All elements that are both in A and in B.



Intersection Examples

Given

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

$$C = \{6, 7, 8\}$$

$$D = \{\} \text{ (Empty set)}$$

$$A \cap B = \{3, 4\}$$

$$A \cap D = \{\}$$

(intersection with an empty set results in the empty set)

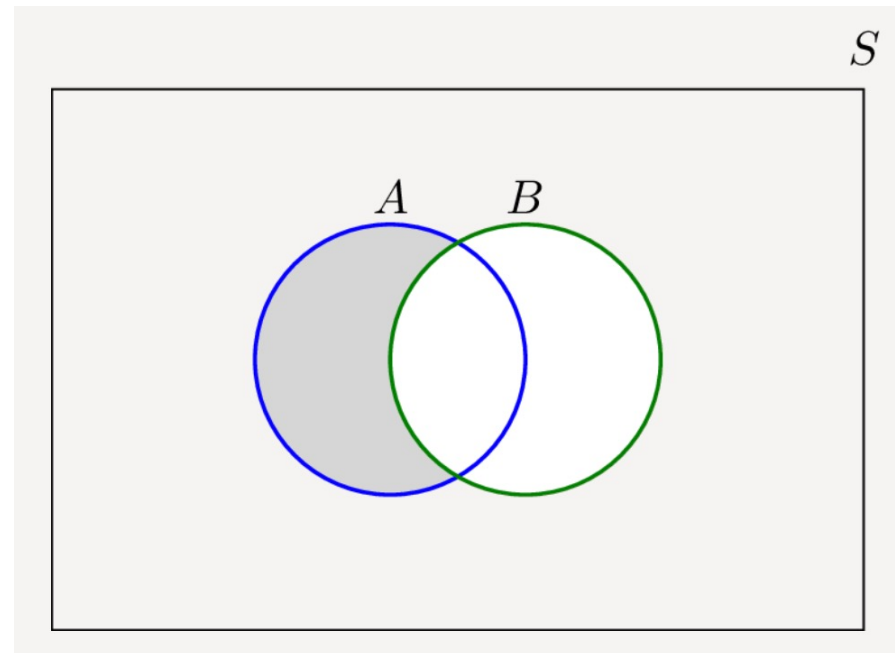
$$B \cap C = \{6\}$$

Set Operations: Difference

$$A \setminus B$$

Create a new set (C) by taking all of the elements of the first set (A) and removing from it all of the elements in the second set (B)

All elements in A but not in B



Difference Examples

Given

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

$$C = \{6, 7, 8\}$$

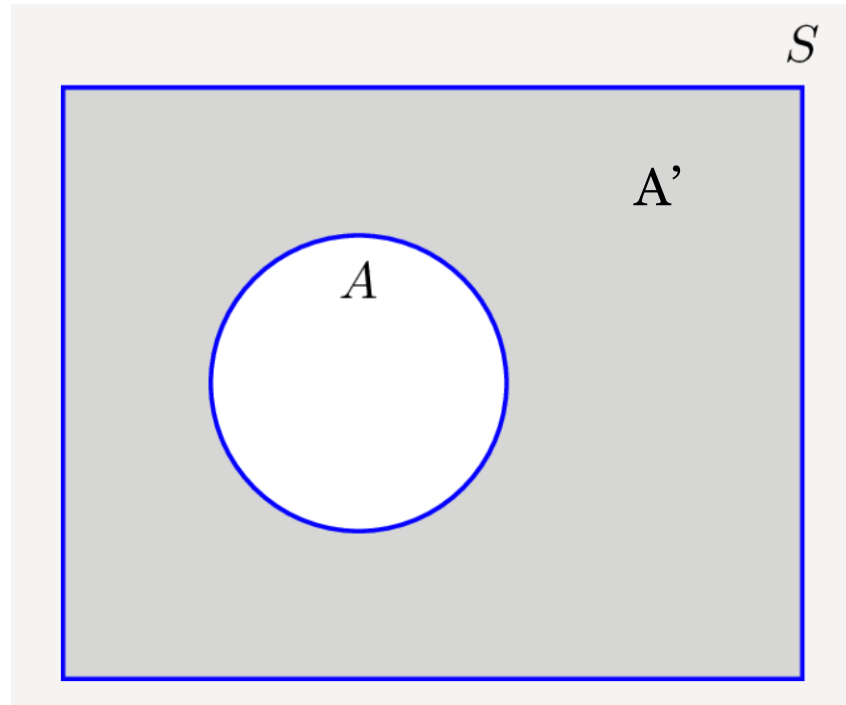
$$D = \{\} \text{ (Empty set)}$$

- $A \setminus B = \{1, 2\}$ (Elements in A that are not in B)
- $A \setminus D = \{1, 2, 3, 4\}$ (Difference with an empty set results in the original set)
- $C \setminus B = \{7, 8\}$

Set Operations: Complementation

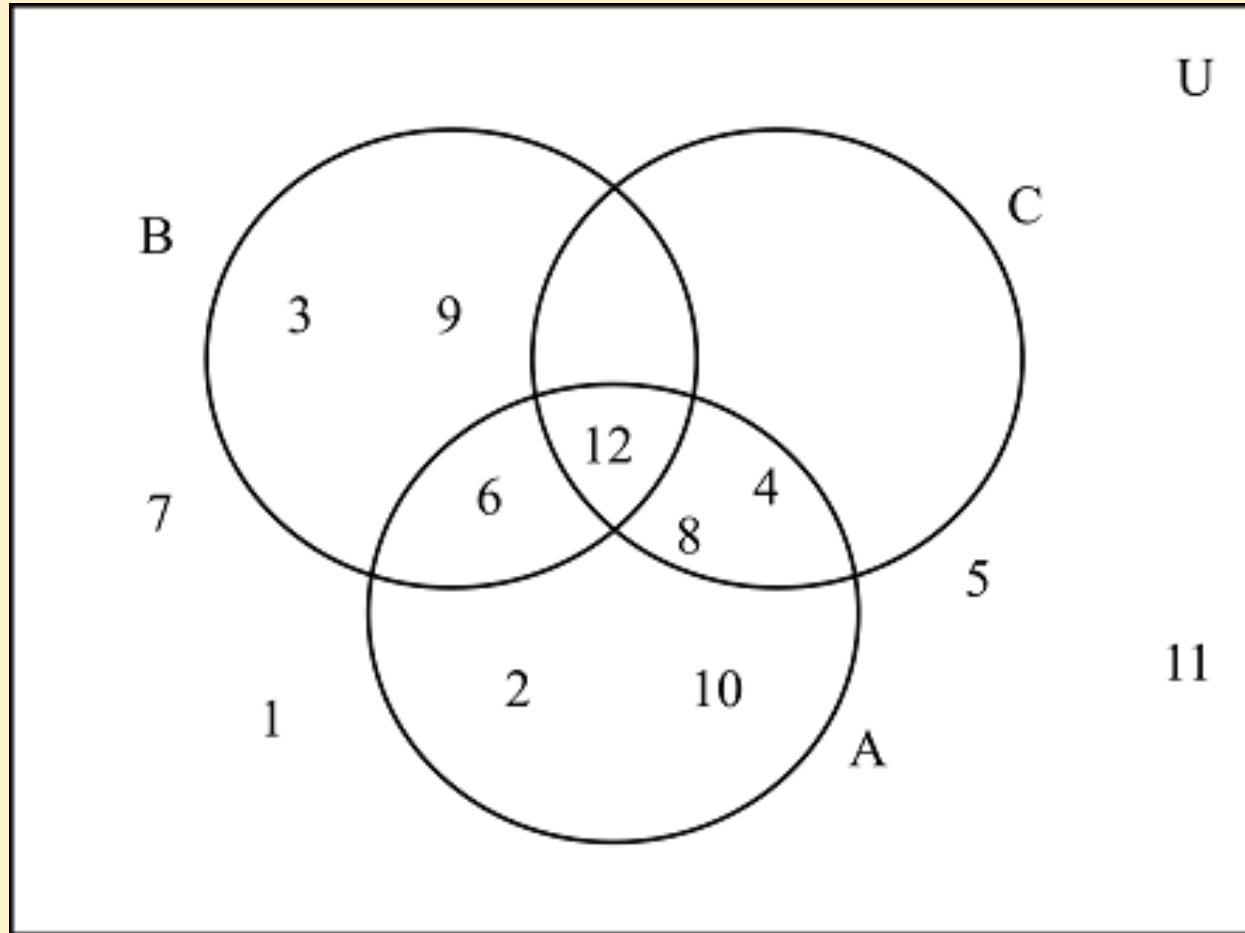
A' or A^c

All elements not in A , with respect to a universal set U (S).



Activity 3:

Name the Elements of the Given Sets



A, B, C,

A',

B/C,

A ∩ C,

A ∪ B ∪ C

Fundamental Laws of Set Algebra

Commutative Laws:

Union: $A \cup B = B \cup A$

Intersection: $A \cap B = B \cap A$

Associative Laws:

Union: $A \cup (B \cap C) = (A \cup B) \cap C$

Intersection: $A \cap (B \cup C) = (A \cap B) \cup C$

Distributive Laws:

Over Union: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Over Intersection: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Identity Laws:

Union: $A \cup \emptyset = A$

Intersection: $A \cap U = A$ (where U is the universal set)

Complement Laws:

$A \cup A' = U$ (where A' is the complement of A and U is the universal set)

$A \cap A' = \emptyset$

Idempotent Laws:

Union: $A \cup A = A$

Intersection: $A \cap A = A$

Double Negation Law:

$(A')' = A$

Absorption Laws:

$A \cup (A \cap B) = A$

$A \cap (A \cup B) = A$

Power Set

The **power set** of a set is the set of all possible subsets of that set, including the *empty set* and the *set itself*.

If S is a set, the power set of S is usually denoted as $\mathbf{P(S)}$ or 2^S .

Example:

For the set $S=\{a,b\}$, the power set $P(S)$ is: $P(S)=\{\emptyset,\{a\},\{b\},\{a,b\}\}$

Cardinality of Power Set: The number of elements in the power set of S is 2^n , where n is the number of elements in S .

For example, $|P(S)|=2^2=4$

Example: Power Set

$$S = \{p, q, r, s\}$$

The power set $P(S)$ is the set of all possible subsets of S , including the empty set and S itself.

$$P(S) = \{\emptyset, \{p\}, \{q\}, \{r\}, \{s\}, \{p, q\}, \{p, r\}, \{p, s\}, \{q, r\}, \{q, s\}, \{r, s\}, \{p, q, r\}, \{p, q, s\}, \{p, r, s\}, \{q, r, s\}, \{p, q, r, s\}\}$$

There are a total of $2^4 = 16$ subsets in the power set $P(S)$

Cartesian Product

The **Cartesian product** of two sets is the set of all ordered pairs where the first element in each pair is from the first set, and the second element is from the second set. The Cartesian product of sets A and B is denoted as $A \times B$.

Example:

$A = \{1, 2\}$ and $B = \{a, b\}$; $A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$

Order Matters!

The order of sets in a Cartesian product is significant. $A \times B$ is not necessarily the same as $B \times A$.

For instance, using the sets from the above example,
 $B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$, which is different from $A \times B$.

Intervals

Intervals Basics

What are Intervals?

Intervals are a subset of *the real number line*, usually referring to all numbers between two endpoints.

The nature of the inclusion of these endpoints defines different types of intervals.

Interval Type	Definition	Description
Open Interval	(a,b)	Contains all numbers between a and b but not including a and b.
Closed Interval	$[a,b]$	Contains all numbers between a and b including a and b.
Half-Open Interval (Left)	$(a,b]$	Contains all numbers between a and b, including b but excluding a.
Half-Open Interval (Right)	$[a,b)$	Contains all numbers between a and b, including a but excluding b.
Infinite Interval (Right)	(a,∞)	Contains all numbers greater than a.
Infinite Interval (Left)	$(-\infty,a)$	Contains all numbers less than a.
Closed Infinite (Right)	$[a,\infty)$	Contains all numbers greater than or equal to a.
Closed Infinite (Left)	$(-\infty,a]$	Contains all numbers less than or equal to a.

Intervals: Graphical Representation

Inequality	Interval Notation	Graph on Number Line	Description
$x > a$	(a, ∞)		x is greater than a
$x < a$	$(-\infty, a)$		x is less than a
$x \geq a$	$[a, \infty)$		x is greater than or equal to a
$x \leq a$	$(-\infty, a]$		x is less than or equal to a

Intervals: Graphical Representation

$a < x < b$	(a, b)		x is strictly between a and b
$a \leq x < b$	$[a, b)$		x is between a and b , to include a
$a < x \leq b$	$(a, b]$		x is between a and b , to include b
$a \leq x \leq b$	$[a, b]$		x is between a and b , to include a and b

Examples of Interval Tasks

Task 1. Determine if the provided number belongs to the given interval, which contains *integers* only:

1. Interval: $(2,5)$ Number: 3.751
2. Interval: $[1,4)$ Number: 1
3. Interval: $(5,8]$ Number: 5
4. Interval: $[-2,2]$ Number: 0
5. Set: $\{-2, 2\}$ Number: 0

Answer:

1. Yes; 2. Yes; 3. No; 4. Yes; 5. No

Task 2. For each set described by the given set builder notation, identify the corresponding interval:

$$A = \{x \mid -2 < x < 4\};$$

$$B = \{x \mid 1 \leq x < 5\};$$

$$C = \{x \mid x \leq 0\}$$

Answer:

$$A = (-2, 4);$$

$$B = [1, 5);$$

$$C = (-\infty, 0]$$

Sets vs Intervals

Attribute	Sets	Intervals
Definition	A collection of distinct objects.	A continuous section of the number line.
Notation	Listed elements, e.g., $\{1,2,3\}$	Bracketed endpoints, e.g., $[a,b]$
Elements	Can be discrete or continuous.	Always continuous (on the real number line).
Examples	$\{2,4,6\}$	$(2,5)$
Inclusivity	Explicitly includes each listed element.	Defined by the type of interval (open, closed, half-open)
Representation	Can represent numbers, characters, etc.	Represents a range of numbers.